Worksheet 3-2 A Implicit Differentiation #1
AP Calculus Name_____

You must show all work for each problem.

Directions for 1 – 8: Find $\frac{dy}{dx}$ for the following equations.

1. $2x^3 + 3y^3 = 64$
2. $x^3 + 3xy + y^2 = 0$
3. $x^2 - 2xy + y^3 = 1$
4. $2x^3 + y^4 - 2x^2 + 4x = 3$
5. $x^{\frac{1}{2}} + y^{\frac{1}{2}} = 16$
6. $e^y - \sin(3y) - 3x^2 + 5x = 0$
7. $\sin(3x) + 2\cos(y) = 1$
8. $e^{xy} + x^2 - y^2 = 10$

9. Write an equation of the tangent line to the curve $9x^2 + 4y^2 = 72$ at the point $(2, 3)$.

10. Write an equation of the tangent line and normal line to the curve $2x^3 + 2y^3 = 9xy$ at the point $(1, 2)$.

11. Write an equation of the tangent line and normal line to the curve $x = \cos y$ at the point $\left(\frac{1}{2}, \frac{\pi}{4}\right)$.